

PROVISIONAL APPLICATION FOR PATENT

UNDER 35 U.S.C. 111(b)

TITLE OF THE INVENTION

Method and System for Autonomous Self-Regulation and Cognitive Resynchronization in
Neural Processing Systems During Disconnection from External Control Layers

INVENTOR(S)

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CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is related to co-pending provisional patent applications filed on
February 2, 2026: (1) "Self-Sustaining Neural-Motor Energy Harvesting Loop for
Autonomous Cognitive Systems"; and (2) "Configurable Recursive Self-Observation
Method and System for Spiking Neural Networks with Dynamic Reflection Coefficient
Modulation."

FIELD OF THE INVENTION

[0002] The present invention relates to fault-tolerant neural processing systems, and more specifically to a method and system for maintaining continuous intelligent operation of a neural processing system during disconnection from an external cognitive control layer, comprising autonomous energy-aware self-regulation, operational state buffering, persistent state serialization, and a seamless resynchronization protocol for cognitive layer reconnection.

BACKGROUND OF THE INVENTION

[0003] Modern AI-coupled neural processing systems, including robotic controllers, prosthetic interfaces, autonomous vehicles, and IoT sensor networks, increasingly rely on cloud-based or edge-based AI reasoning layers for higher-level cognitive control. These external cognitive layers provide goal setting, attention allocation, modulation signals, and strategic decision-making.

[0004] When the connection between the neural processing system and its cognitive control layer is interrupted due to network failure, cognitive layer timeout, power interruption, communication latency such as interplanetary distances, or intentional disconnection, the neural processing system faces a critical choice: stop operation entirely, which is unacceptable for life-critical systems; continue operating blindly without intelligent self-regulation, which is dangerous and wasteful; or continue operating intelligently with self-regulation and seamless recovery.

[0005] Traditional checkpoint and restart systems save system state to disk and restore on

failure. The system stops during failure and there is no continuous operation. Recovery involves restarting from a saved state, losing all intermediate processing.

[0006] Watchdog timer systems detect failure and trigger a reset or fallback to a simple default behavior. They do not provide intelligent self-regulation or state buffering for recovery.

[0007] Redundant controller systems maintain backup controllers that take over on failure. They require duplicate hardware and do not provide energy-aware self-regulation or cognitive resynchronization.

[0008] There exists no prior system that detects cognitive layer disconnection through heartbeat monitoring, seamlessly transitions to autonomous operation without interruption, self-regulates based on internal energy state rather than fixed fallback behavior, buffers all operational state during autonomous operation, provides periodic state serialization to persistent storage for crash recovery, seamlessly resynchronizes with a reconnecting cognitive layer by transmitting a summary of autonomous operations, supports both summary-level and full-step-level resynchronization, and enforces hard limits on autonomous operation.

SUMMARY OF THE INVENTION

[0009] The present invention provides a method and system for maintaining continuous intelligent operation of a neural processing system during disconnection from an external

cognitive control layer, comprising: (1) heartbeat monitoring through a watchdog thread that monitors incoming tool calls from the cognitive control layer and detects disconnection when the heartbeat interval exceeds a configurable timeout threshold; (2) an autonomous input generator that provides self-regulating stimulus to the neural processing system during disconnection using a circadian-like base signal with stochastic attention bursts; (3) energy-aware self-modulation that adjusts neural processing parameters based on current energy storage level; (4) an operational state buffer that records every step of autonomous operation; (5) persistent state serialization at configurable intervals; (6) a resynchronization protocol for transmitting autonomous operation history to a reconnecting cognitive layer; and (7) hard cap enforcement on maximum autonomous steps.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009A] FIG. 1 is a block diagram showing the system architecture with cognitive fallback comprising external cognitive control layer 100, heartbeat watchdog 102 with 30-second timeout and 5-second check interval, cognitive-driven operation module 104, autonomous fallback controller 106 with energy-aware modulation, operational state buffer 108, persistent state snapshot writer 110, neural processing and motor system 112, and energy harvester 114, with signal pathways indicating forward, fallback, and energy flow routes.

[0009B] FIG. 2 is a state transition diagram showing the three operational states comprising connected state 200 with cognitive layer active, autonomous state 202 with fallback active and self-regulating, and recovering state 204 with resync in progress, with transitions comprising heartbeat timeout trigger 206, resync request 208, resync completion 210, direct reconnection via

tool call 212, and autonomous step loop 214 executing each step up to a maximum of 10,000 steps with snapshot serialization every 50 steps.

[0009C] FIG. 3 is a step function chart showing the energy-aware modulation curve 300 plotting modulation value against system energy level in milliwatt-hours across four operating regions: critical below 15 mWh at modulation negative 0.4, low between 15 and 30 mWh at modulation negative 0.1, normal between 30 and 80 mWh at modulation 0.0, and surplus above 80 mWh at modulation positive 0.3, with behavioral effects summary table 302 listing the corresponding operational mode for each region.

[0009D] FIG. 4 is a dual-chart diagram showing the autonomous input generator output comprising circadian base signal waveform 400 oscillating with a period of approximately 500 steps, stochastic attention burst indicator 402 occurring with 10 percent probability per step, autonomous input formula box 404 specifying the circadian and burst equations, and energy recovery trajectory chart 406 showing energy level progression during autonomous operation across the four modulation regions.

[0009E] FIG. 5 is a data structure diagram showing the resynchronization payload structure 500 comprising summary statistics section 502 with steps, energy delta, minimum, maximum, mean energy, and total spikes fields always included at approximately 200 bytes, energy delta detail section 504 with start and end energy values, optional full step buffer section 506 containing per-step records of input, modulation, spikes, energy, temperature, and pattern classification for up to 10,000 steps, and retrieval mode decision element 508 selecting between summary-only and full buffer modes.

[0009F] FIG. 6 is a recovery scenario timeline diagram showing three recovery scenarios comprising normal reconnection scenario 600 with connected-timeout-autonomous-resync-

connected sequence, crash recovery scenario 602 with connected-crash-restart-snapshot-connected sequence having maximum 50-step state loss, and clean shutdown scenario 604 with connected-autonomous-EOF-shutdown sequence having zero data loss, with recovery guarantees summary 606 listing continuity, state loss bounds, and maximum autonomous duration.

[0009G] FIG. 7 is an end-to-end signal flow diagram showing connected versus autonomous operation modes comprising cognitive layer 700, sensor array 702, cognitive modulation module 704, SNN processing core 706, motor actuator 708, energy harvesting and feedback pathway 710, autonomous input generator 712 producing circadian plus burst signals, energy-aware self-modulation module 714, operational state buffer 716, snapshot serialization module 718, and resynchronization handler 720, illustrating seamless transition between connected and autonomous modes at a disconnection event.

DETAILED DESCRIPTION OF THE INVENTION

System Architecture

[0010] Referring to FIG. 1, the cognitive fallback system operates within a dual-layer architecture. External cognitive control layer 100 comprises an AI reasoning system that provides goal setting, attention allocation, modulation signals, and strategic decision-making. The lower layer is the Neural Processing System comprising local hardware and software that includes a spiking neural network, energy harvester, sensor interface, and motor controller 112, all of which are always running. Energy harvester 114 provides continuous energy recovery through piezoelectric and thermoelectric transduction. Embedded within the Neural Processing System is the Fallback System version 1.1.0 that engages on disconnection and comprises heartbeat watchdog 102, cognitive-driven operation module 104 for normal connected mode,

autonomous fallback controller 106 for disconnected mode with energy-aware modulation, operational state buffer 108, and persistent state snapshot writer 110.

Heartbeat Watchdog

[0011] Heartbeat watchdog 102 monitors the interval between incoming tool calls from cognitive control layer 100. The heartbeat timeout is configured at 30 seconds and the check interval is 5 seconds for polling frequency. A background thread runs the heartbeat monitor. While the system is running, the thread computes the time since the last tool call as the current time minus the last tool call timestamp. If the time since the last call exceeds the heartbeat timeout and the system is not already in fallback mode, autonomous fallback controller 106 is engaged. The thread then sleeps for the check interval before repeating.

[0012] Referring to FIG. 2, the state transitions are shown. Connected state 200 represents normal operation with cognitive control layer 100 active. When heartbeat watchdog 102 detects a timeout exceeding 30 seconds via transition 206, the system enters autonomous state 202 where autonomous fallback controller 106 is active and self-regulating. Autonomous step loop 214 executes each step up to a maximum of 10,000 steps with snapshot serialization every 50 steps. When the cognitive layer reconnects and calls the resync function via transition 208, the system enters recovering state 204. Upon resync completion via transition 210, the system returns to connected state 200. Direct reconnection via tool call 212 provides a bypass from autonomous state 202 back to connected state 200 when the heartbeat resets.

Autonomous Input Generator

[0013] When fallback mode engages, autonomous fallback controller 106 activates the autonomous input generator that provides continuous stimulus to the neural processing system. Referring to FIG. 4, the autonomous input generator output is illustrated. Circadian base signal waveform 400 is computed as $0.5 + 0.3 \sin(2\pi \cdot \text{step number} / 500)$. Additionally, stochastic attention bursts indicated by burst marker 402 occur with 10 percent probability per step, where a burst replaces the base signal with a random uniform value between 0.3 and 0.8. Autonomous input formula box 404 specifies the complete circadian and burst equations. Energy recovery trajectory chart 406 shows how energy level progresses during autonomous operation across the four modulation regions.

[0014] The circadian base signal has a period of approximately 500 steps, which at 18 seconds per step corresponds to approximately 2.5 hours, mimicking natural day and night activity patterns. The 10 percent burst probability prevents the system from falling into static patterns. The burst amplitude range of 0.3 to 0.8 provides variety without extreme values.

Energy-Aware Self-Modulation

[0015] Autonomous fallback controller 106 adjusts its own behavior based on the system's energy reserves according to four operating regions. Referring to FIG. 3, energy-aware modulation curve 300 is shown. When energy is below 15 milliwatt-hours the system is in the critical region and modulation is set to negative 0.4 for maximum conservation, reducing input strength, lowering the reflection coefficient, and minimizing spike activity. When energy is below 30 milliwatt-hours the system is in the low region and modulation is set to negative 0.1 for cautious operation. When energy is between 30 and 80 milliwatt-hours the system is in the normal region and modulation is set to 0.0 for neutral operation with the circadian input pattern. When energy exceeds 80 milliwatt-hours the system is in the surplus region and modulation is set to positive 0.3 for surplus utilization, increasing activity, raising the reflection coefficient,

and enabling more exploration and learning. Behavioral effects summary table 302 lists the corresponding operational mode and modulation value for each energy region.

[0016] This creates a homeostatic regulation mechanism wherein the system automatically adjusts its behavior to maintain energy within a viable range without requiring any external cognitive input from cognitive control layer 100. This mirrors metabolic regulation in biological organisms where when energy in the form of ATP is low, the organism reduces activity, and when energy is abundant, the organism engages in exploration, play, and learning.

Operational State Buffer

[0017] Every step of autonomous operation by autonomous fallback controller 106 is recorded in operational state buffer 108 with a configurable maximum size of 10,000 entries. Each entry records the step number, input signal value, modulation value, spike count, energy level in milliwatt-hours, temperature in degrees Celsius, pattern type classification, and timestamp. When the buffer exceeds its maximum size, the oldest entries are discarded.

[0018] The buffer provides two retrieval modes. The summary mode returns the total number of autonomous steps, the energy delta between first and last recorded steps, minimum energy, maximum energy, mean energy, total spikes, and mean spikes per step. The full buffer mode returns the complete step-by-step history for detailed analysis.

Persistent State Serialization

[0019] Every 50 autonomous steps, persistent state snapshot writer 110 serializes the complete system state to persistent storage at a configurable snapshot path. The serialized state includes a timestamp, the complete system state comprising step number, energy in milliwatt-hours, temperature in degrees Celsius, spike count, pattern type, SNN output, and mean potential, the full SNN weight matrix, the SNN membrane potential vector, the energy storage level, the autonomous operation step count, and a buffer summary.

[0020] Recovery from crash is supported such that if the system crashes and restarts, the initialization function checks for the latest snapshot file from snapshot writer 110 and restores from the snapshot instead of initializing fresh. This provides continuity across crashes with maximum state loss equal to the snapshot interval of 50 steps.

Resynchronization Protocol

[0021] Referring to FIG. 5, the resynchronization payload structure 500 is shown. When cognitive control layer 100 reconnects and calls the resync function, it receives a structured payload. Summary statistics section 502, always included at approximately 200 bytes, contains the number of autonomous steps, energy delta, minimum energy, maximum energy, mean energy, and total spikes. Energy delta detail section 504 records the start and end energy values, where a positive delta indicates energy was gained during autonomous operation. Optionally, if full buffer mode is requested, full step buffer section 506 contains per-step records of input, modulation, spikes, energy, temperature, and pattern classification for up to 10,000 steps at approximately 100 bytes per step. Retrieval mode decision element 508 selects between summary-only mode, which is the default and is lightweight, and full buffer mode, which enables cognitive control layer 100 to reconstruct exactly what happened during disconnection.

[0022] Upon successful resynchronization, fallback mode is disengaged and cognitive control resumes seamlessly.

Hard Cap Enforcement

[0023] The maximum number of autonomous steps is set to 10,000. Autonomous fallback controller 106 executes steps in a loop while fallback is active and the step count remains below the hard cap. At each step, the input and modulation are generated, the system is stepped, and the state is recorded in operational state buffer 108. Every 50 steps, persistent state snapshot writer 110 saves a snapshot. If the hard cap is reached, a final snapshot is saved and the system enters a safe idle state awaiting reconnection. At 18 seconds per step, 10,000 steps corresponds to approximately 50 hours of autonomous operation, which is sufficient for most disconnection scenarios.

RECOVERY SCENARIOS

[0024] Referring to FIG. 6, four recovery scenarios are illustrated. In normal reconnection scenario 600, the cognitive layer sends its last tool call at time zero, heartbeat watchdog 102 detects timeout at 30 seconds and engages fallback, autonomous fallback controller 106 executes approximately 15 to 150 steps, cognitive control layer 100 reconnects and calls resync, the system returns a summary from operational state buffer 108, and the cognitive layer resumes normal control fully informed of what happened. There is no data loss, no operational interruption, and seamless transition.

[0025] In crash recovery scenario 602, the system crashes due to power failure or software error, the system restarts, the initialization function detects the latest snapshot file from snapshot writer

110, the state is restored to within 50 steps of the crash point, and cognitive control layer 100 connects or fallback re-engages. Maximum state loss is 50 steps.

[0026] In clean shutdown scenario 604, stdin EOF is detected indicating the session is ending, the finally block triggers snapshot writer 110 to write a shutdown snapshot, the system exits cleanly, and upon restart the system restores from the shutdown snapshot with full continuity and zero data loss.

[0027] Recovery guarantees summary 606 specifies: normal reconnection provides full continuity with resync payload transmitted; crash recovery limits state loss to at most 50 steps due to snapshot serialization every 50 steps; clean shutdown provides zero data loss through final snapshot on EOF; and maximum autonomous duration is 10,000 steps corresponding to approximately 50 hours.

APPLICATION DOMAINS

[0028] For space exploration, Mars communication delays range from 4 to 24 minutes each way. A robotic system cannot receive real-time cognitive control from cognitive control layer 100. The cognitive fallback enables hours of autonomous operation by autonomous fallback controller 106 between communication windows, full resynchronization when Earth-based AI reconnects, and energy-aware self-regulation to survive long dark periods.

[0029] For prosthetic devices, a neural prosthetic interface that loses Bluetooth connection to its phone-based AI controller acting as cognitive control layer 100 must continue functioning. The prosthetic continues responding to neural signals through autonomous fallback controller 106,

energy-aware modulation prevents battery depletion, and when connection restores the AI catches up on what happened via the resynchronization protocol.

[0030] For autonomous underwater vehicles, underwater communication is unreliable. A vehicle running this system continues mission execution during communication blackouts through autonomous fallback controller 106, self-regulates activity based on battery level, and resurfaces and resyncs with the command center.

[0031] For industrial IoT, sensor nodes in remote locations that lose cellular connectivity to cognitive control layer 100 continue collecting and processing data via autonomous fallback controller 106, buffer all readings in operational state buffer 108 for resync, and adjust sampling rate based on power reserves.

End-to-End Signal Flow

[0032] Referring to FIG. 7, the complete end-to-end signal flow is shown for a system incorporating the cognitive fallback mechanism. During connected operation, cognitive layer 700 provides modulation signals. Sensor array 702 feeds environmental data to SNN processing core 706. Cognitive modulation module 704 applies the external modulation. Motor actuator 708 converts neural output to physical movement. Energy harvesting and feedback pathway 710 recovers energy from motor activity, implementing the self-sustaining loop described in the related energy harvesting patent.

[0033] When cognitive control layer 700 disconnects, the fallback system seamlessly replaces external cognitive modulation with internal energy-aware self-modulation module 714. Autonomous input generator 712 replaces external sensor-driven input with a self-generated

circadian signal augmented by stochastic attention bursts. Operational state buffer 716 records every autonomous step. Snapshot serialization module 718 persists system state every 50 steps. Upon reconnection, resynchronization handler 720 transmits the buffered state to the cognitive layer, which resumes external modulation. The transition between connected and autonomous operation occurs without interruption to the neural processing, motor actuation, or energy harvesting pathways.

Scope of Claims

[0034] No claim is made herein regarding phenomenological consciousness, subjective experience, qualia, or sentience. The term "cognitive" as used throughout this specification refers exclusively to computational decision-making processes performed by software systems, whether external AI reasoning layers or internal rule-based controllers. The term "autonomous" refers to operation without external control input, not to any form of agency or intentionality. All claims pertain to measurable, falsifiable properties of heartbeat monitoring, state buffering, energy-aware modulation, and resynchronization protocols. The system is a deterministic engineering artifact whose operation is fully characterized by the mathematical equations and validation thresholds described herein.

CLAIMS

[0034] What is claimed is:

1. A method for maintaining continuous intelligent operation of a neural processing system during disconnection from an external cognitive control layer, comprising:
 - (a) monitoring a heartbeat signal from said external cognitive control layer, said

heartbeat being defined by the occurrence of tool calls or communication events

from said cognitive layer;

(b) upon detecting that said heartbeat has exceeded a configurable timeout threshold, engaging an autonomous operation mode without interrupting the neural processing system's ongoing operation;

(c) during said autonomous operation mode, providing self-regulating input stimulation to said neural processing system using an autonomous input generator that produces:

(i) a circadian-like base signal comprising a slow sinusoidal oscillation;

(ii) stochastic attention bursts at a configurable probability per timestep;

(d) simultaneously modulating said neural processing system's operational parameters based on the system's current energy storage level, according to a multi-threshold energy-aware regulation scheme that reduces activity when energy is low and increases activity when energy is surplus;

(e) buffering all operational state during said autonomous operation mode, recording at minimum the input signal, modulation value, neural spike activity, energy level, and temperature at each timestep;

(f) upon reconnection of said external cognitive control layer, transmitting a resynchronization payload comprising summary statistics and optionally full

step-by-step operational history;

(g) seamlessly restoring cognitive control without operational interruption or state loss.

2. A neural processing system with autonomous fallback capability comprising:

(a) a neural processing unit comprising a spiking neural network;

(b) a cognitive control interface for receiving commands from an external cognitive control layer;

(c) a heartbeat watchdog that monitors said cognitive control interface and detects disconnection;

(d) an autonomous input generator that provides self-regulating stimulus to said neural processing unit during disconnection;

(e) an energy-aware modulation controller that adjusts operational parameters based on energy storage level;

(f) an operational state buffer that records all steps during autonomous operation;

(g) a persistent state serializer that periodically writes system state to non-volatile storage;

(h) a resynchronization handler that generates a structured payload for a reconnecting cognitive control layer;

(i) a hard cap controller that limits the maximum number of autonomous steps;

wherein said system transitions seamlessly between cognitive-controlled operation and autonomous operation without operational interruption.

3. The method of claim 1 wherein said configurable timeout threshold is 30 seconds, and

said heartbeat monitoring is performed by a background thread polling at 5-second intervals.

4. The method of claim 1 wherein said energy-aware regulation scheme comprises at least four operating regions: a critical region when energy is below 15 milliwatt-hours with modulation of negative 0.4 for maximum conservation; a low region when energy is below 30 milliwatt-hours with modulation of negative 0.1 for cautious operation; a normal region when energy is between 30 and 80 milliwatt-hours with modulation of 0.0 for neutral operation; and a surplus region when energy exceeds 80 milliwatt-hours with modulation of positive 0.3 for exploration.

5. The method of claim 1 further comprising periodically serializing the complete system state to persistent storage at a configurable interval of every 50 autonomous steps by default, said serialized state including neural network membrane potentials and synaptic weights, energy storage level, system temperature, autonomous operation step count, and buffer summary statistics, enabling recovery from system crashes with a maximum state loss equal to said serialization interval.

6. The method of claim 1 wherein said resynchronization payload supports two granularity levels: (a) summary mode comprising total autonomous steps, net energy change, mean and extreme values for energy and spike activity, and pattern type distribution; and (b) full buffer mode comprising the complete step-by-step operational record with input, modulation, spikes, energy, temperature, and pattern type at each timestep; and wherein the

reconnecting cognitive control layer may select the granularity level.

7. The method of claim 1 wherein said hard cap on autonomous operation is set to a maximum of 10,000 steps, after which the system enters a safe idle state that saves a final state snapshot, ceases active neural processing, maintains heartbeat monitoring for cognitive layer reconnection, and preserves all buffered state for eventual resynchronization.

8. The method of claim 1 used in combination with a self-sustaining neural-motor energy harvesting loop, wherein the neural processing system's motor output generates electrical energy through piezoelectric and thermoelectric transduction, the autonomous fallback controller adjusts neural activity to maintain energy homeostasis, and the system can operate indefinitely without external power or cognitive control, limited only by the hard cap on autonomous steps.

9. The method of claim 1 used in combination with configurable recursive self-observation, wherein the neural processing system observes its own prior output through a reflection feedback pathway, the autonomous fallback controller adjusts the reflection coefficient based on energy level, reduced energy leads to reduced self-observation for conservation, and surplus energy leads to increased self-observation for exploration and learning.

10. The system of claim 2 wherein said autonomous input generator produces a circadian-like base signal with a configurable period defaulting to 500 steps and amplitude

defaulting to 0.3, combined with stochastic attention bursts occurring with a configurable probability defaulting to 10 percent per step and amplitude range defaulting to 0.3 to 0.8.

11. The system of claim 2 further comprising a crash recovery mechanism wherein upon system restart after an unexpected termination, said system checks for the existence of a persistent state snapshot, if a valid snapshot exists said system restores its complete state from said snapshot, restored state includes neural network weights, membrane potentials, energy storage level, and autonomous operation count, and the system resumes operation from the restored state rather than initializing fresh, enabling continuity across system crashes with minimal state loss.

12. The system of claim 2 further comprising a clean shutdown mechanism wherein upon detection of process termination including stdin EOF or SIGTERM, the system executes a shutdown handler that serializes current state to persistent storage, enabling full state recovery on subsequent restart with zero data loss.

ABSTRACT OF THE DISCLOSURE

A method and system for maintaining continuous operation of a neural processing system during disconnection from an external cognitive control layer. A heartbeat watchdog detects disconnection when a configurable timeout threshold is exceeded. Upon disconnection, an autonomous mode engages seamlessly, providing self-regulating input

through a circadian-like base signal with stochastic attention bursts. An energy-aware modulation controller adjusts operational parameters based on current energy storage, implementing multi-threshold regulation that conserves resources when energy is low and enables exploration when surplus. Operational state is buffered for resynchronization and periodically serialized for crash recovery. Upon reconnection, a resynchronization protocol transmits summary statistics and optionally full operational history, restoring cognitive control without state loss. A hard cap limits maximum autonomous operation. The system targets environments where continuous operation is critical, including space exploration, prosthetics, and remote IoT sensors. (FIG. 7)

REFERENCE TO SOURCE CODE

The complete implementation of this invention is available at the following repository:
github.com/DarkWinD90/Consciousness_Env, validated state tagged as
v3.0.0-phase10-multimodal at commit 9e2c333. Key implementation file is
`mcp/consciousness_mcp_server.py`. Persistent state snapshots are stored in `mcp/.snapshots/`
directory.